

WIRELESS NETWORKING

DAY – 1 Introduction to networking and the OSI model theory.

Task – 1 Wireshark Installation .

- When capturing traffic on your network interface while visiting a website like `instagram.com`, Wireshark records every layer of communication. Common Protocol Names Seen in the Packet List :-
- DNS (Domain Name System): Resolves human-readable domain names (e.g., `instagram.com`) into IP addresses before connecting. TCP (Transmission Control Protocol): Manages connection setup (3-way handshake), reliable data transmission, and connection teardown.
- TLS / SSL (Transport Layer Security): Handles the cryptographic handshake (Client Hello, Server Hello) and secures the encrypted tunnel used by modern HTTPS sites.
- QUIC / HTTP/3: A modern UDP-based transport protocol used heavily by platforms like Meta (Instagram) and Google to speed up web traffic. ARP (Address Resolution Protocol): Resolves local IP addresses to physical MAC addresses on your local network/router.
- SSDP / MDNS: Local service discovery protocols continuously broadcasting on home or local networks.
- Typical Order of Protocols Observed
- DNS Query & Response $\rightarrow$ Resolves `instagram.com` to its target IP address. TCP Handshake $\rightarrow$ SYN, SYN-ACK, ACK packets to establish the socket connection.

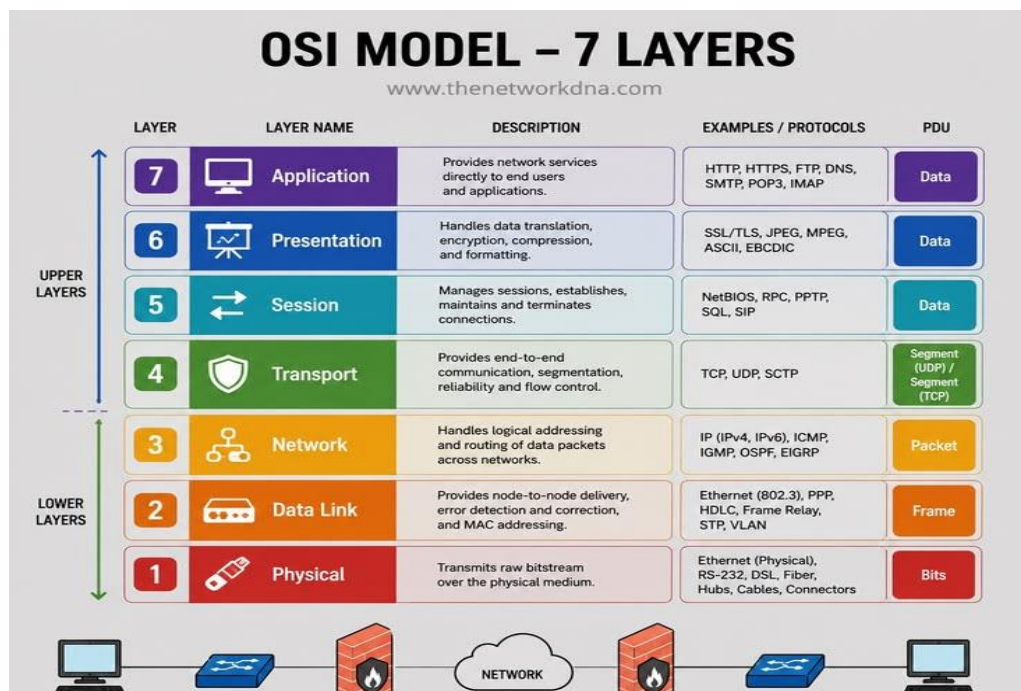
- TLS Handshake $\rightarrow$ Exchanging encryption keys to establish a secure session. TLS Application Data / HTTP/3 $\rightarrow$ Encrypted website content, media, and images streaming to your browser.

Task – 2 Packets from my wireshark and finding which OSI layer in packet mainly represents.

- Packet Analysis & OSI Layer Mapping :-
- Packet 1: DNS (Domain Name System)
- OSI Layer: Application Layer (Layer 7) :- DNS operates at the top of the OSI model. It takes human-readable domain names (like `instagram.com`) and translates them into machine-readable IP addresses for user applications like web browsers.
- Packet 2: TCP (Transmission Control Protocol)
- OSI Layer: Transport Layer (Layer 4) :- TCP manages end-to-end communication, session establishment (via SYN, SYN-ACK, ACK), flow control, and error recovery between your laptop and the web server.
- Packet 3: ARP (Address Resolution Protocol)
- OSI Layer: Data Link Layer (Layer 2) :- ARP maps network layer IP addresses to physical media access control (MAC) addresses, enabling devices on the exact same local Ethernet or Wi-Fi network segment to frame and deliver data to each other.

Task – 3 OSI diagram

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Task – 4 How msg travels on application.

- Sending a WhatsApp message involves a complex network journey. When you type and send a text, it moves down the OSI layers on your phone, across the network as electronic/light signals, and back up the OSI layers on the recipient's phone (or WhatsApp's server).
- Here is how sending a WhatsApp message travels through 4 key layers of the OSI Model:
 - **1. Application Layer (Layer 7)**
 - What happens: This is the top layer where your interaction begins. When you type a message in the WhatsApp user interface and tap the "Send" button, the application formats your text, timestamps it, and applies end-to-end encryption to protect your privacy. WhatsApp prepares the actual raw content of your message (the text/emoji) and turns it into data that the networking stack can process securely.
 - **2. Transport Layer (Layer 4)**
 - Before traveling over the internet, your message needs a reliable way to reach its destination without data loss. At Layer 4, protocols like TCP (Transmission Control Protocol) slice your encrypted message into manageable chunks called segments. Each segment is assigned a port number (identifying it specifically as WhatsApp traffic) and a sequence number so the receiving device can reconstruct the message in the correct order and request missing pieces if any are lost. This layer acts like a quality-assurance manager, breaking the message into trackable packages and making sure every single piece arrives intact.
 - **3. Network Layer (Layer 3)**
 - Once the segments are ready, Layer 3 attaches source and destination IP addresses to them, creating what are called IP packets. This layer is responsible for logical addressing and routing—determining the best path through routers and intermediate networks across the internet to reach WhatsApp's servers and eventually the recipient's device. Think of this as putting a physical postal address on an envelope so routers on the internet know where to direct the packet.

- 4. Physical Layer (Layer 1)
- At the very bottom, all the structured digital data (0s and 1s) must be converted into physical signals so it can leave your device. Depending on your connection, your phone's hardware translates these bits into radio frequency waves (for Wi-Fi or cellular networks) or light/electrical impulses (if passing through fiber-optic cables or copper lines along the internet backbone). This layer turns digital binary data into actual physical signals (light, radio waves, or electricity) that travel through cables or the air to transfer your message.

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